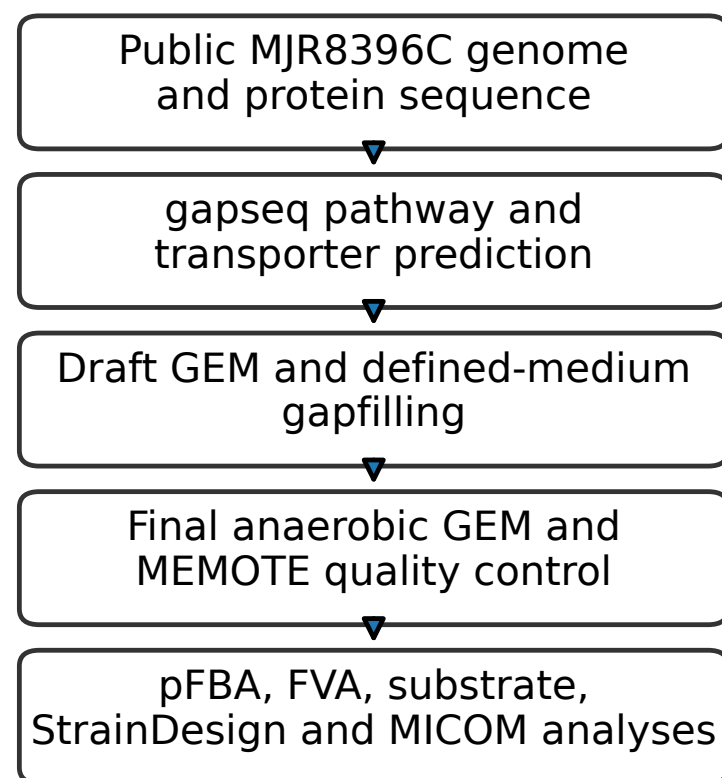
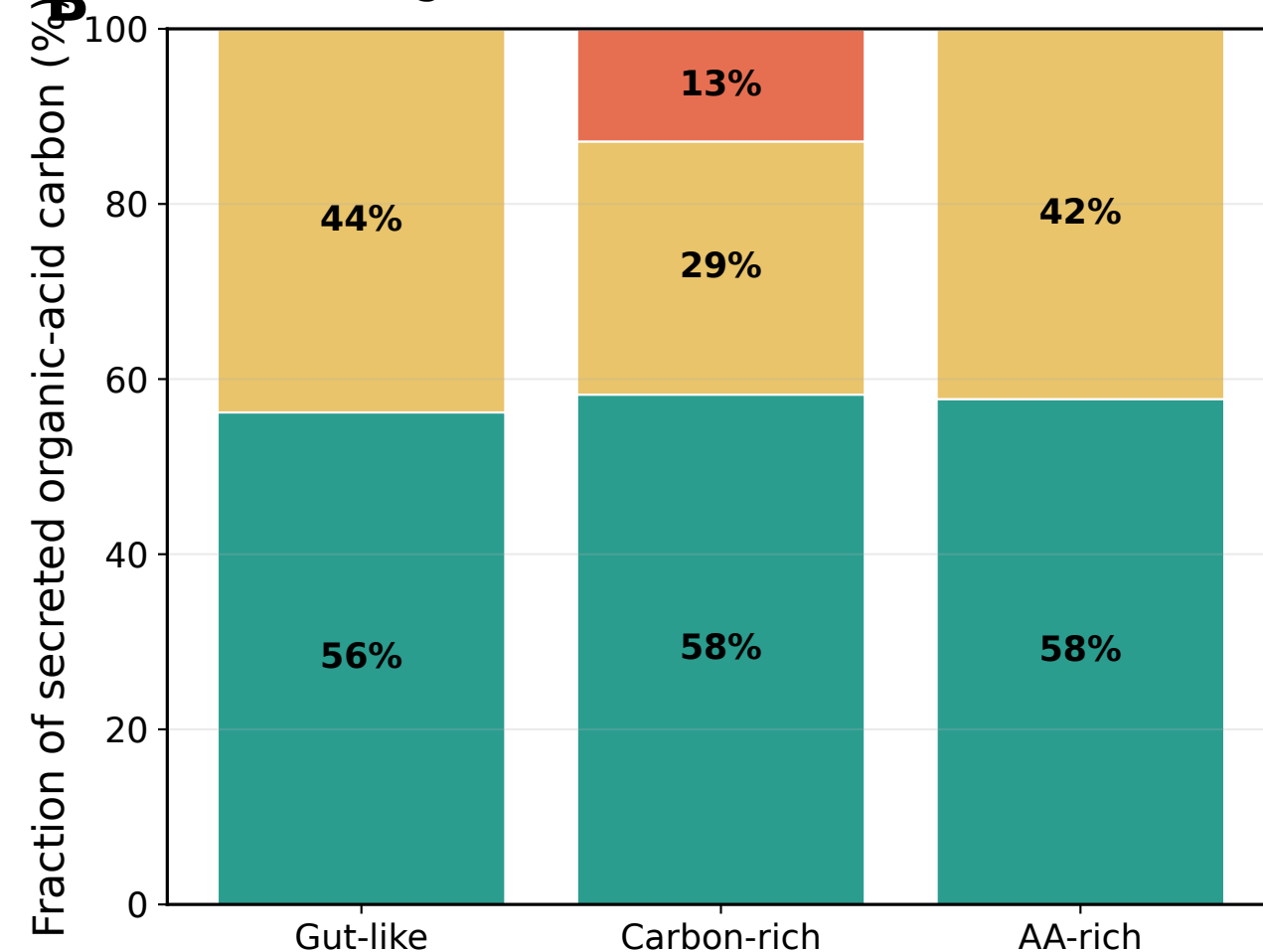


**A**

## Reconstruction and analysis workflow

**B**

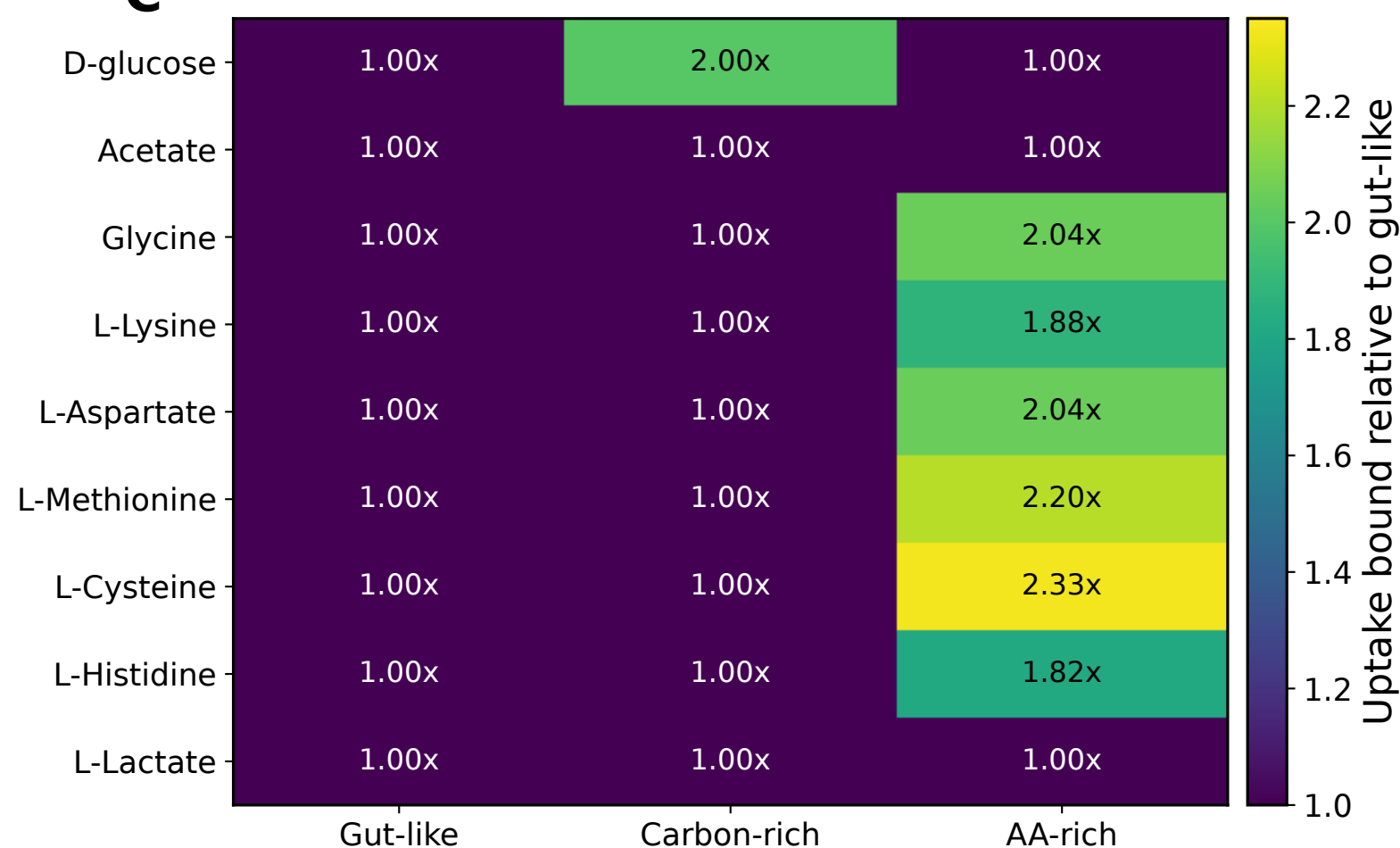
## Organic-acid carbon allocation



Butyrate      Formate  
Propionate      Succinate

**C**

## Controlled medium perturbations

**D**

## Growth versus secreted organic-acid carbon

